WRS

for

THEIA: Indoor Navigation App for the Blind and Visually Impaired

Version 1.4

Prepared by Ubuntu Bros.

Reviewed by All

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1. Introduction

1.1 Purpose

The purpose of this project is to develop software that will allow users with vision impairment to have an easier time safely navigating multi-floor environments. Although many buildings provide many services to make buildings more accessible, indoor navigation remains a significant problem due to inconsistent building layouts, dynamic obstacles, and lack of knowledge on indoor positioning.

THEIA aims to address this by combining real-time obstacle detection, indoor positioning data, and accessible interactions such as voice commands, audio cues, and haptic feedback. The system is designed to operate under uncertainty to adapt to the changes of the indoor environments, as well as provide safe fallbacks when positioning data becomes unavailable. The goal is to create a navigation tool that improves safety, independence, and confidence for blind and visually impaired users in indoor spaces.

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1.2 Scope

THEIA will focus on indoor navigation within multi-floor buildings such as academic facilities, offices, and public institutions. The system will use available indoor positioning when present (e.g. floor maps or Wi-Fi triangulation) but is also designed to function when such infrastructure is incomplete or unreliable. The scope of the project includes:

- Deteccting Obstacles and environmental changes in real time
- Providing navigation instructions through audio cues
- Supporting hands-free interaction through voice commands
- Guiding users safely when positioning data is lost
- Allowing caregivers to configure the system when needed
- Ensuring privacy through consent-based data sharing.

1.3 Objectives and Success Criteria

- Provide reliable indoor navigation instructions using audio and haptic phone functions
- Support hands-free use through voice commands
- Detect and anounce obstacles
- Safely handle positioning failures by guiding users to last known point
- Operate effectively in dynamic indoor environments
- Maintain user privacy through consent-based data sharing
- Be intuitive and non obstructive for blind and visually imparied users

1.4 Definitions, Acronyms, and Abbreviations

- **THEIA –** Indoor navigation application developed in this project
- **Indoor Positioning –** Techniques for locating a user inside a building
- **Audio Cues –** Spoken instructions or alerts provided by the system
- **Haptic Feedback –** Vibrations used to communicate navigation or alerts
- **Domain –** The environment and conditions in which THEIA operates
- **FR –** Functional Requirements
- **NFR –** Non-Functional Requirement

1.5 Overview

This document will document the issues with the preliminary definitions and explain the improved understanding of those definitions. Section 2 will go over the Issues with the given Preliminary Definition including issues with the domain, stakeholders, functional and non-functional objectives. Section 3 will identify the problems and goals as well as explain the improved understanding of the issues presented in section 2. Section 4 will provide mockups of the preliminary prototype as well as user manual.

2. Issues with Preliminary Definition Given (ambiguities, incompleteness, inconsistency, conflicts, ...)

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2.1 Issues with II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

2.1.1 Issue DI-1: Indoor positioning is assumed but never stated

Issue description: The domain assumes the building has a way to locate the user indoors (beacons, Wi-Fi, or digital floor maps). This assumption is never written down. If a building lacks this infrastructure, or it fails, THEIA cannot navigate, and the definition doesn't say what happens.

Type: incompleteness

Options:

1. Leave it implicit and assume all target buildings have positioning.
2. State it as an explicit domain assumption only.
3. State it as an explicit domain assumption and define fallback behavior when positioning is lost.

Decision and rationale: Option 3. Blind users depend on THEIA for safety, so losing positions without warning is dangerous. Stating the assumption makes the dependency visible, and a fallback (i.e announcing "location unavailable" and guiding the user back to the last known point) keeps the user safe when it fails.

2.1.2 Issue DI-2: Hands-free use conflicts with how the app is described

Issue description: The stakeholders include users with canes or service animals who don't have both hands free. The objectives still describe a phone app the user holds and taps.

Type: conflict

Options:

1. Keep touch-based interaction and expect users to stop walking to use the app.
2. Make hands-free operation a core objective: voice commands, haptic and audio cues, and use with the phone in a pocket or on a lanyard.
3. Add voice commands as an optional extra.

Decision and rationale: Option 2. Users with mobility aids are a primary user group, not an edge case. Making them stop and hold the phone would make the app impractical and unsafe while walking.

2.1.3 Issue DI-3: The indoor environment is treated as static

Issue description: The domain describes rooms, stairwells, and elevators as fixed. Real indoor spaces change constantly: people move through hallways, furniture is rearranged, doors close, and areas get blocked for maintenance.

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Type: incorrect domain assumption

Options:

1. Rely only on stored floor maps.
2. Rely on floor maps and let building staff update them manually.
3. Use floor maps for routes, add real-time obstacle detection, and warn the user when map data may be outdated.

Decision and rationale: Option 3. Floor maps are good for planning a route but can't show what's in the hallway right now. Since obstacle awareness is a core user need, the app must handle changes it can't predict.

2.1.4 Issue DI-4: Caregivers are listed as stakeholders but have no objective

Issue description: Caregivers and family members are named as people who set up the app and come to the user's aid. No functional objective covers setup by a caregiver or contacting them. Sharing a user's location with others also raises privacy concerns that aren't addressed.

Type: incompleteness

Options:

1. Remove caregivers from the stakeholder list.
2. Add caregiver setup only.
3. Add caregiver setup and optional emergency alerts, with the user's consent required before any location is shared.

Decision and rationale: Option 3. Caregivers clearly have a role in the user's safety. Requiring consent protects the user's independence and privacy while still letting them get help.

2.2 Issues with II.2 Software System Requirements: Functional Requirements

2.2.1 Issue FI-1: User goals and system behavior are mixed together

Issue description: "Help identify indoor surroundings and obstacles" combines what the user wants to achieve (reach a destination safely) with what the system does (detect and announce obstacles). Mixing the two makes the requirement hard to test and hard to trace. (Type: ambiguity, requirement and specification mixed)

Options:

1. Keep the combined statement.
2. Keep only the user goal and leave system behavior undefined.

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3. Split it into a requirement (user reaches the destination safely without sight) and a specification (system announces obstacles within a set distance through audio or haptics).

Decision and rationale: Option 3. Separating the requirement from the specification lets each one be checked independently, and it fits the WRS model the project follows.

2.2.2 Issue FI-2: No requirement covers moving between floors

Issue description: The domain is defined as multi-floor buildings with stairwells and elevators, but no functional requirement covers navigating from one floor to another. The app cannot meet its goal in the domain it targets. (Type: incompleteness)

Options:

1. Limit THEIA to single-floor navigation.
2. Support elevators only.
3. Support both elevators and stairwells and let the user choose: detect the current floor, guide to the elevator or stairwell, and confirm arrival on the new floor.

Decision and rationale: Option 3. Option 1 drops a core use case. Some users can't use stairs and others prefer them, so the choice should be the user's.

2.3 Issues with II.3 Software System Requirements: Non-Functional Requirements

2.3.1 Issue NI-1: "Easily discernible UI" cannot be tested

Issue description: The requirement that users with partial perception get an "easily discernible" UI is subjective. There is no way to verify whether it has been met. (Type: ambiguity, unverifiable)

Options:

1. Keep the wording and judge it informally.
2. Verify it through user feedback only.
3. Replace it with measurable criteria: high-contrast mode meeting WCAG 2.1 AA, adjustable text size, and full compatibility with screen readers (VoiceOver and TalkBack).

Decision and rationale: Option 3. Measurable criteria can be tested and checked against a recognized accessibility standard. User feedback can still be gathered to confirm them.

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2.3.2 Issue NI-2: A reliable network connection is assumed

Issue description: The requirements assume the app always has a network connection. Indoor signal is often weak, especially in stairwells and elevators, which are exactly where navigation matters most. (Type: incompleteness, hidden assumption)

Options:

1. Require a connection and show an error when it drops.
2. Cache only the current route.
3. Store building maps on the device so core navigation works offline, and use the network only for updates.

Decision and rationale: Option 3. A blind user losing guidance mid-stairwell is a safety risk. Offline core navigation keeps the app dependable in the places it's needed most.

3. WRS

3.1 W

3.1.1 Problem

Blind and visually impaired individuals face significant challenges when trying to navigate indoors independently. Compared to outdoors, where GPS is widely available and easy to use, indoors doesn't have the same level of accessible navigation information. Building layouts are much more complex and compact, including multiple floors, stairwells, and elevators, making it difficult for users to determine their location.

The following problems have been identified to be core to the app:

Problem ID	Problem Name	Problem Description
P1	Limited Indoor Positioning	Indoor environments lack uniform structures, making it hard for blind users to navigate unfamiliar buildings
P2	Difficulty Navigating Multiple Floors	Buildings with multiple floors, stairwells, and elevators create more navigation challenges. Users may get stuck/be unable to navigate depending on which floor they are on
P3	Unreliable Network Connectivity	Indoor environments may have spotty or inaccessible network connectivity
P4	Inaccessible User Interface	Traditional user interfaces are not accessible to blind and visually impaired users, who

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		need other forms of communication to use an app
P5	Independence for Blind and Visually Impaired Users	Blind and visually impaired users should be able to explore on their own, while not having to worry about emergency scenarios, as well as be able to build up their own routine

3.1.2 Goals

Goal ID	Goal Name	Goal Description
G1	Reliable Indoor Navigation	Blind and visually impaired users should be able to determine their location and navigate indoors safely
G2	Safe Navigation in Dynamic Environments	Buildings with multiple floors, stairwells, and elevators should be able to be recognized and routed carefully based on accessibility for the user
G3	Offline Navigation	Navigation should be available offline for users through previously saved maps
G4	Accessible User Interface	User interface should be high contrast, as well as connect to screen readers
G5	Achieve independence for Blind and Visually Impaired users	Blind and visually impaired users assisted through the collection of daily routine data, then suggesting next actions based off previous data, as well as contacting emergency services in cases of user request and non-responsiveness after a duration set by the user

3.1.3 Improved understanding of II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

3.1.3.1 Improved Domain

Improved Domain ID	Improved Domain Description
ID1	Indoor multi-floor settings, containing multiple rooms, stairwells, elevators
ID2	Indoor positioning infrastructure required for navigation, use of digital floor maps, Bluetooth beacons, and Wi-Fi triangulation. Not guaranteed across all buildings(DI-1)
ID3	Indoor environments are dynamic: People and furniture move, doors open/close, and obstructions may be temporary. (DI-3)

3.1.3.2 Stakeholders

- Sponsors
  - Faculty: Bolong Zeng
- Potential Users:
  - Fully Blind/No light perception users will use the application to help identify indoor surroundings and obstacles these users will not be able to interact with visual UI elements in a traditional way.

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- Users with Partial Perception may use the app and need a mix of UI elements and haptic feedback, easily discernable UI could be helpful
- Users with mobility aids like service animals or canes will need to interact with the application without full use of both hands.
- Caregivers/Family members that will configure the app for others and come to their aid, this will require consent from the Blind Users to allow position sharing in emergencies.(DI-4)
- Accessibility Department Staff Members who would like students to use an app like THEIA

3.1.3.3 Improved Functional Objectives

Improved FR Objective ID	Objective Description	Alleviates Problems	Achieved Goals
FO-1	**The system should safely guide users through indoor environments, including between multiple floors**	**P1, P2**	**G1, G2**
FO-2	**The system should include fallbacks in case of emergencies/loss of services, including contacting emergency services and offline core navigation**	**P3, P5**	**G3, G5**

3.1.3.4 Improved Non-Functional Objectives

Improved NFR Objective ID	Objective Description	Alleviates Problems	Achieved Goals
NFO-1	**The system should use high-contrast mode meeting WCAG 2.1 AA, adjustable text size, and full compatibility with screen readers (VoiceOver and TalkBack)**	**P4**	**G4**
NFO-2	**The system should use offline navigation as its core functionality**	**P3**	**G3**

3.2 RS

3.2.1 Functional RS – Improved understanding of II.2 Software System Requirements: FRs

Improved FR ID	Functional Requirement Description	Alleviates Problems	Achieved Goals
FR-1	**The system shall allow users to specify an indoor**	**P1, P4, P5**	**G1, G4, G5**

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	destination through voice commands or user interface controls		
FR-2	**The system shall determine the user's current indoor location using available positing information and provide navigation instructions toward the selected destination**	**P1, P2**	**G1**
FR-3	**The system shall calculate available routes to a destination and allow users to select the route based on their accessibility needs and preferences**	**P1, P2**	**G1, G2**
FR-4	**The system shall provide step-by-step navigation instructions using audio cues and directions, announcing arrival at destination**	**P2, P4**	**G4**
FR-5	**The system shall detect obstacles using available sensors and notify users through audio or haptic feedback**	**P2, P4**	**G2, G4**
FR-6	**The system shall support navigation through multiple floors by providing directions to stairwells or elevators based on user preferences**	**P1, P2**	**G1, G2**
FR-7	**The system shall provide an ability to notify emergency services if the user requests, or is unresponsive after a set duration**	**P5**	**G5**
FR-8	**The system shall record the users daily common routes, as well as have the ability to accept imports of daily schedules, and suggest and predict next actions**	**P5**	**G5**

3.2.2 Non-functional RS -Improved understanding of II.2 Software System Requirements: NFRs

Improved NFR ID	Functional Requirement Description	Alleviates Problems	Achieved Goals
NFR-1	**The system shall provide a high-contrast visual interface that meets applicable WCAG 2.1 AA accessibility requirements**	**P4**	**G4**

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NFR-2	**The system shall support adjustable text sizes and compatibility with mobile screen readers**	**P4**	**G4**
NFR-3	**The system shall provide essential navigation instructions through audio and haptic feedback so users can navigate without visual information**	**P1, P2, P4**	**G1, G2, G4**
NFR-4	**The system shall provide core navigation functionality without a continuous network connection by using locally stored navigation data**	**P3**	**G3**
NFR-5	**The system shall allow users to customize audio volume, instruction frequency, and navigation preferences according to their individual accessibility needs**	**P2, P4**	**G2, G4**
NFR-6	**The system shall remain responsive to essential user interactions, including emergency assistance requests and offline navigation**	**P3, P5**	**G3, G5**

4. Preliminary Prototype and User Manual

4.1 Prototype overview

The preliminary prototype is a rough, small scale project of the THEIA mobile app for IOS and Android. It shows the main screens and how the user moves between them. It does not include real indoor positioning. Its purpose is to check the improved requirements with stakeholders such as caregivers and accessibility staff before detailed design begins.

The Prototype follows three design rules from the improved requirements:

- The voice and haptics: this is where every actions can be done without looking at the screen (D1-2).
- Large Touch Targets: On the main screens, the whole screen acts as one button, so the user doesn't have to find a small button(D1-2,N1-1).
- Works offline: Building maps are stored on the device incase of no service (N1-2).

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4.2 Prototype screen

Screen View	Purpose	Main interactions	Traces
S1 Home	Starting point	Double tap any where on screen so say, “Hey THEIA” App will announce users’ location	FR-1 FR-2
S2 Choose Destination	Pick where to go	Say the room name, swipe through rooms, saved and suggested places. App repeats the choice to confirm	FR-1 FR-8
S3 Route Options	Choose a route	App reads out the available routes. User picks one based on their needs. Options like elevator only, avoid stairs)	FR-3 FR-6
S4 Arrival	Confirms destination	Announces arrival with a vibration type setting or a voice alert. offers to save as favorite	FR-4

4.2.1 Vibration Patterns Used in the prototype

Pattern	Meaning
1 short pulse	Turn left
2 short pulses	Turn right
1 long pulse	Obstacle ahead, stop
3 quick pulses	Arrived at destination
Pulse every few seconds	Approaching elevators or stairs

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4.3: User Manual

4.3.1 Getting Started

- Install THEIA from the App Store or Google Play
- Allow access to location, microphone and motion sensors when asked
- Download the map for the building. Maps are stored on the phone so it can be accessible when offline.
- If Voiceover or Talkback is used THEIA works with to automatically

4.3.2 Going to a Destination

- Double tap anywhere on the screen or say “Hey THEIA”
- Say where you want to go, for example locate chinook building, location is saved and can be accessed on the app
- app will repeat destination. Say “yes” or double tap to confirm
- follow the spoken directions and vibrations, phone can be in pocket or backpack
- THEIA lets you know that you have arrived.

4.3.3 Changing floors

If your destination is on another floor, THEIA asks whether you want the elevator or the stairs. It guides you there, then confirms when you reach the new floor before continuing.

4.3.4 Obstacle Warning

When THEIA detects something in your path, it vibrates with one long pulse and says what is ahead. If the building map may be out of date, THEIA tells you to proceed with caution.

4.3.5 Hands-Free Use

If you use a cane or service animal, all actions work by voice. Say “Repeat” to hear

The last direction, for instance user uses commands like “where am I to know what there current location or Stop to end navigation.

4.3.6 Settings

Open settings to change the speed of the speech, Vibration level, and Text Size.

4.3.7 Caregiver Setup

A caregiver can add an emergency contact in the profile setup, Location of the User is never shared unless approve.

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4.3.8 Emergency Help

For emergency User can hold volume button for 3 seconds. App alerts emergency contacts and automatically shares location.

4.3.9 Troubleshooting

Problem	What to do
Location Unavailable	User stands still for a few seconds so app can find position
No Service	Navigation will keep working with saved map. Map should be able to reconnect as soon as service is restored
Wrong Direction	Say “stop” then start the route again

5. Traceability

5.1. Forward Traceability Matrix

Problem ID	Goal ID	Functional Objectives	Non-Functional Objectives	Functional Requirements	Non-Functional Requirements
P1	**G1**	**FO-1**	-	**FR-1, FR-2, FR-3, FR-4**	**NFR-3**
P2	**G2**	**FO-1**	-	**FR-2, FR-3, FR-5, FR-6**	**NFR-3, NFR-5**
P3	**G3**	**FO-2**	**NFO-2**	**-**	**NFR-4, NFR-6**
P4	**G4**	**-**	**NFO-1**	**FR-1, FR-4, FR-5**	**NFR-1, NFR-2, NFR-3, NFR-5**
P5	**G5**	**FO-2**	**-**	**FR-1, FR-7, FR-8**	**NFR-6**

5.2. Backward Traceability Matrix

Requirement ID	Related Objectives	Related Problems	Related Goals
FR-1	**FO-1**	P1, P4, P5	**G1, G4, G5**
FR-2	**FO-1**	P1, P2	**G1**
FR-3	**FO-1**	**P1, P2**	**G1, G2**
FR-4	**FO-1**	**P1, P4**	**G1, G4**
FR-5	**FO-1**	**P2, P4**	**G2, G4**
FR-6	**FO-1**	**P2**	**G2**
FR-7	**FO-2**	**P5**	**G5**
FR-8	**FO-2**	**P5**	**G5**
NFR-1	**NFO-1**	**P4**	**G4**
NFR-2	**NFO-1**	**P4**	**G4**
NFR-3	**NFO-1**	**P2, P4**	**G2, G4**

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NFR-4	**NFO-2, FO-2**	**P3**	**G3**
NFR-5	**NFO-1**	**P2, P4**	**G2, G4**
NFR-6	**FO-2, NFO-2**	**P3. P5**	**G3, G5**

The traceability matrices demonstrate that each of the five identifies problems and goals are addressed by at least one functional or non-functional system requirement.

Reference

WCAG 2 Overview | Web Accessibility Initiative (WAI) | W3C

Accessibility | Apple Developer Documentation

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Domain: The extent to which the project will cover

Functional Requirement: Specific, testable description of what the system will do

Non-Functional Requirement: defines how a system must perform

User Manual: Guide on how to use the product

Prototype: Overview of what the product is, how it will work, and how it fits in with our requirements

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